Deterministic Optimization

Discrete Optimization:

Modeling w/ binary variables

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Network Problems

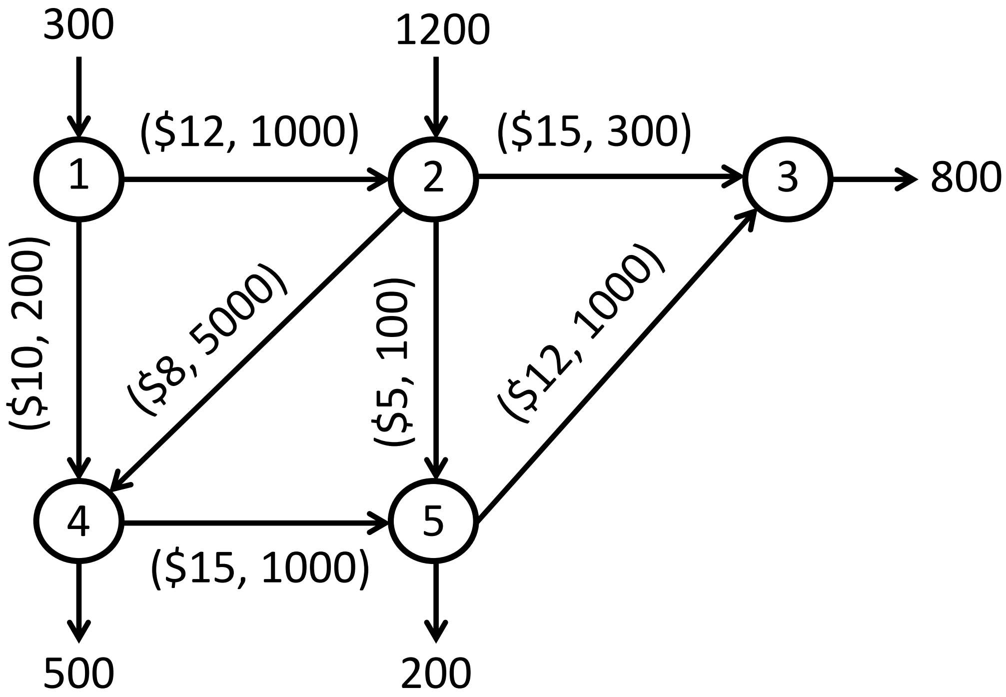
Network Problems

Learning Objectives

- Discover minimum cost network flow problems
- Observe an example using PuLP

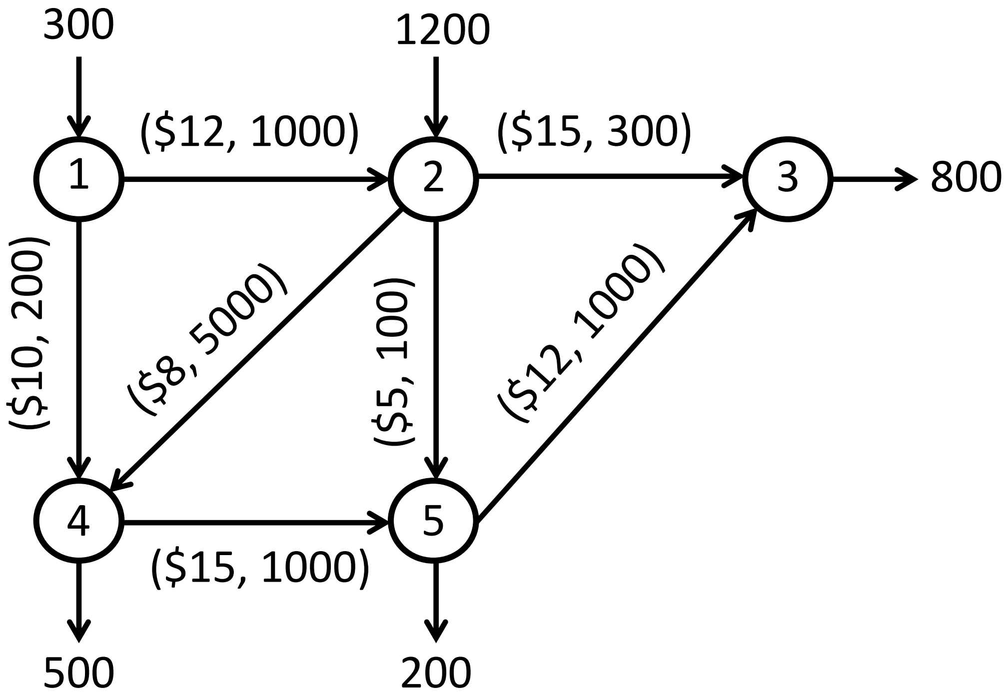
Example

- A company has two manufacturing plants and 3 distribution centers (nodes)
- The plants and DCs are connected by transportation channels (arcs)
- The production amount and demand at each node is shown
- The capacity and cost per unit for each channel (arc) is shown
- Find the cheapest way to move units from plants to DCs



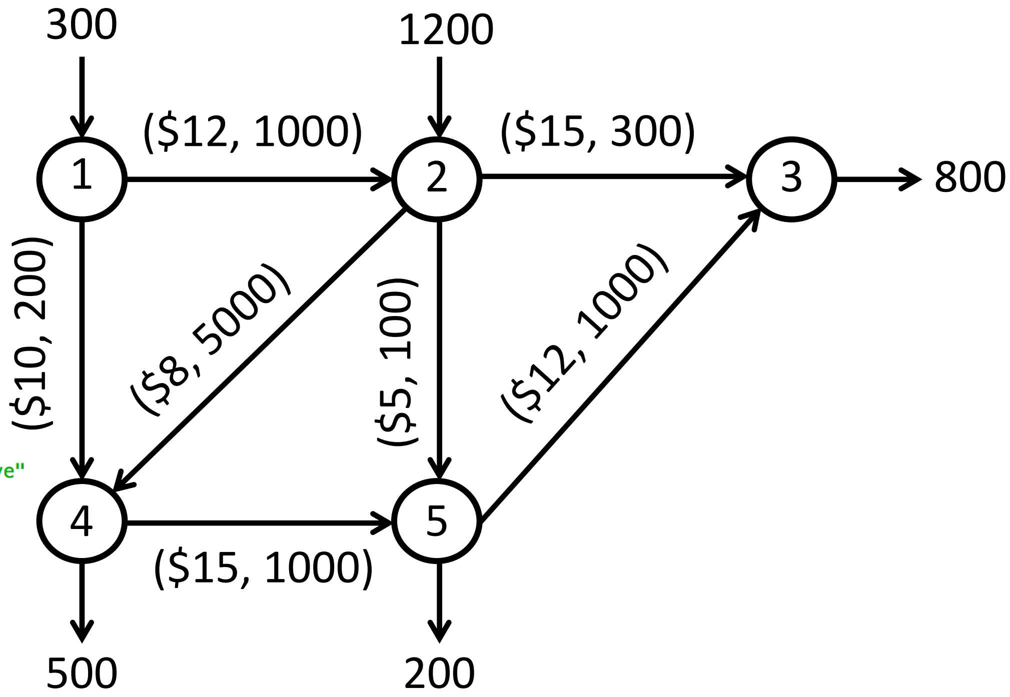
Model

$$\begin{aligned}
\min \quad & 12x_{12} + 10x_{14} + 15x_{23} + 8x_{24} \\
& +5x_{25} + 15x_{45} + 12x_{53} \\
\text{s.t.} \quad & x_{12} + x_{14} = 300 \\
& x_{23} + x_{24} + x_{25} - x_{12} = 1200 \\
& -x_{23} - x_{53} = -800 \\
& x_{45} - x_{14} - x_{24} = -500 \\
& x_{53} - x_{25} - x_{45} = -200 \\
& 0 \le x_{12} \le 1000,\ \ 0 \le x_{14} \le 200, \\
& 0 \le x_{23} \le 300, 0 \le x_{24} \le 5000 \\
& 0 \le x_{25} \le 100,\ \ 0 \le x_{45} \le 1000 \\
& 0 \le x_{53} \le 1000
\end{aligned}$$



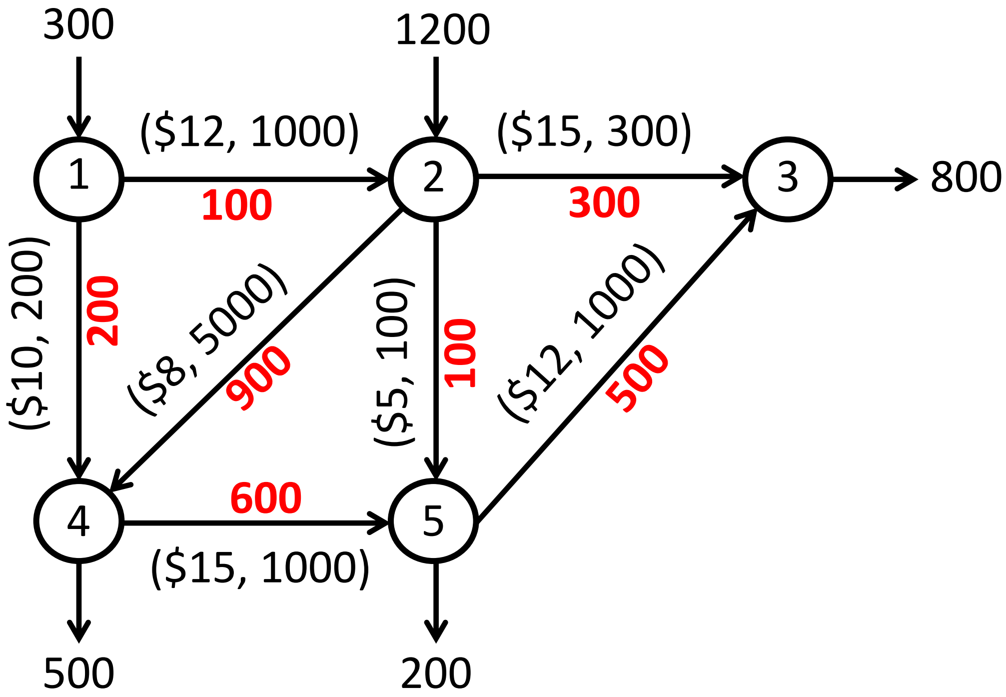
PuLP Code

```python
# Network Optimization
# Install PuLP and glpk before running this code
# http://pythonhosted.org/PuLP/
# https://www.gnu.org/software/glpk/

import pulp as p

# create a model
m = p.LpProblem("Network Model",p.LpMinimize)

# Create decision variables
x12 = p.LpVariable('x12',0,1000,p.LpContinuous)
x14 = p.LpVariable('x14',0,200,p.LpContinuous)
x23 = p.LpVariable('x23',0,300,p.LpContinuous)
x24 = p.LpVariable('x24',0,5000,p.LpContinuous)
x25 = p.LpVariable('x25',0,100,p.LpContinuous)
x45 = p.LpVariable('x45',0,1000,p.LpContinuous)
x53 = p.LpVariable('x53',0,1000,p.LpContinuous)


# Add objective and constraints
m += 12*x12 + 10*x14 + 15*x23 + 8*x24 + 5*x25 + 15*x45 + 12*x53, "Objective"
m += x12 + x14 == 300, "Node 1"
m += x23 + x24 + x25 - x12 == 1200, "Node 2"
m += -x23 - x53 == -800, "Node 3"
m += x45 - x14 - x24 == -500, "Node 4"
m += x53 - x25 - x45 == -200, "Node 5"

m.solve()
for v in m.variables():
    print(v.name, "=", v.varValue)
```



Solution

```
('x12', '=', 100.0)
('x14', '=', 200.0)
('x23', '=', 300.0)
('x24', '=', 900.0)
('x25', '=', 100.0)
('x45', '=', 600.0)
('x53', '=', 500.0)
```



Minimum Cost Network Flow Problem

- We are given a directed network $G = (N, A)$ with a set of nodes N and a set of arcs A.
- Each node $i \in N$ has an associated "supply" b_i. If $b_i > 0$ the node is a supply node, if $b_i < 0$ it is a demand node. We will assume that the network is balanced, i.e. $\sum_{i \in N} b_i = 0$.
- Each arc $(i, j) \in A$ has an associated cost c_{ij} and capacity u_{ij}.
- A flow on this network is a set of values on the arcs that obey capacities and satisfy flow conservation at the nodes.
- The goal is to find a flow of minimum cost.

Minimum Cost Network Flow Problem

$$\begin{aligned}
\min \quad & \sum_{(ij)\in A} c_{ij}x_{ij} \\
\text{s.t.} \quad & \sum_{(ij)\in A} x_{ij} - \sum_{(ji)\in A} x_{ji} = b_i \quad && \forall\ i \in N \\
& 0 \leq x_{ij} \leq u_{ij} && \forall\ (ij) \in A
\end{aligned}$$

Characteristics

- The constraint matrix has $+1$ and -1 in every column and all other entries are 0. Such a matrix is known as a network matrix.
- If the supply data b_i and the capacities u_{ij} are integers, then every extreme point (or basic feasible solution) will have integer entries.
- Thus a network flow problem with integrality requirements can be solved as a linear program.
- This is an important class of integer programs that are "easy"
- Next, we will see some integer programs that can be converted in to network flow problems, and hence can be solved easily.

Shortest Path Problem

Given a graph $G = (N, A)$ with a start node s and destination node t, where each arc (ij) is of length $c_{ij} \geq 0$, find the shortest path from s to t.

$$\begin{aligned}
\min \quad & \sum_{(ij)\in A} c_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_{(ij)\in A} x_{ij} - \sum_{(ji)\in A} x_{ji} = \begin{cases} +1 & i = s \\ 0 & i \in N \setminus \{s, t\} \\ -1 & i = t \end{cases} \\
& x_{ij} \in \{0, 1\} \quad \forall\, (ij) \in A
\end{aligned}$$

Shortest Path Problem

Given a graph $G = (N, A)$ with a start node s and destination node t, where each arc (ij) is of length $c_{ij} \geq 0$, find the shortest path from s to t.

$$\min \quad \sum_{(ij)\in A} c_{ij} x_{ij}$$

$$\text{s.t.} \quad \sum_{(ij)\in A} x_{ij} - \sum_{(ji)\in A} x_{ji} = \begin{cases} +1 & i = s \\ 0 & i \in N \setminus \{s,t\} \\ -1 & i = t \end{cases}$$

$$\cancel{x_{ij} \in \{0,1\} \quad \forall\, (ij) \in A} \qquad 0 \leq x_{ij} \leq 1 \quad \forall (ij) \in A$$

Assignment Problem

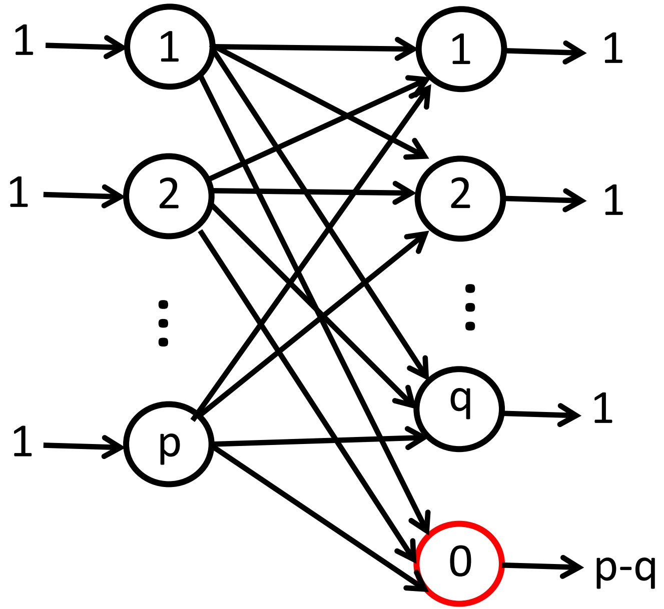
We are given a set of p workers and q tasks. Assume $p > q$. Worker $i \in \{1, \ldots, p\}$ charges \$ c_{ij} to complete task $j \in \{1, \ldots, q\}$. Each worker can be assigned to at most one task. Each task has to be assigned to a worker. Find the minimum cost assignment of worker to tasks.

$$
\begin{aligned}
\min \quad & \sum_{i=1}^{p} \sum_{j=1}^{q} c_{ij} x_{ij} & \\
\text{s.t.} \quad & \sum_{j=1}^{q} x_{ij} \leq 1 & \forall\ i \\
& \sum_{i=1}^{p} x_{ij} = 1 & \forall\ j \\
& x_{ij} \in \{0, 1\} & \forall\ i, j
\end{aligned}
$$

Assignment Problem

The assignment problem can be converted to a min cost network flow problem as follows:

- Each worker is a supply node with supply of 1
- Each task is a demand node with demand of 1
- Introduce a "slack" node with demand of $p - q$
- There is a arc from worker node i to task node j with capacity of 1 and cost of c_{ij}
- There is a arc from each worker node to the slack node with capacity of 1 and cost of 0



Summary

- Network problems are an easy class of integer programs
- Some important classes of integer programs can be converted to network problems